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# King County Metro Knowledge Graph

Transforming Static GTFS Data into Dynamic Visualizations

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1. Information Story
2. What Existed
3. What We Did
4. What Exists Now

# Existing Information

## Current GTFS File

```
agency_id,route_id,route_short_name,route_long_name,route_type,route_desc,route_url,route_color,route_text_color,route_sort_order
1,100001,1,,3,Kinnear - Downtown Seattle,https://kingcounty.gov/en/dept/metro/routes-and-service/schedules-and-maps/001.html,FDB71A,000000,
1,100002,10,,3,Capitol Hill - Downtown Seattle,https://kingcounty.gov/en/dept/metro/routes-and-service/schedules-and-maps/010.html,FDB71A,000000,
1,100003,101,,3,Renton Transit Center - Downtown Seattle,https://kingcounty.gov/en/dept/metro/routes-and-service/schedules-and-maps/101.html,FDB71A,000000,
1,100004,105,,3,Renton Highlands - Renton Transit Center,https://kingcounty.gov/en/dept/metro/routes-and-service/schedules-and-maps/105.html,FDB71A,000000,
1,100005,106,,3,Renton Transit Center - Skyway - Downtown Seattle,https://kingcounty.gov/en/dept/metro/routes-and-service/schedules-and-maps/106.html,FDB71A,000000,
1,100006,107,,3,Renton Transit Center - Rainier Beach,https://kingcounty.gov/en/dept/metro/routes-and-service/schedules-and-maps/107.html,FDB71A,000000,
```

## Map PDFs



routes.txt ¶

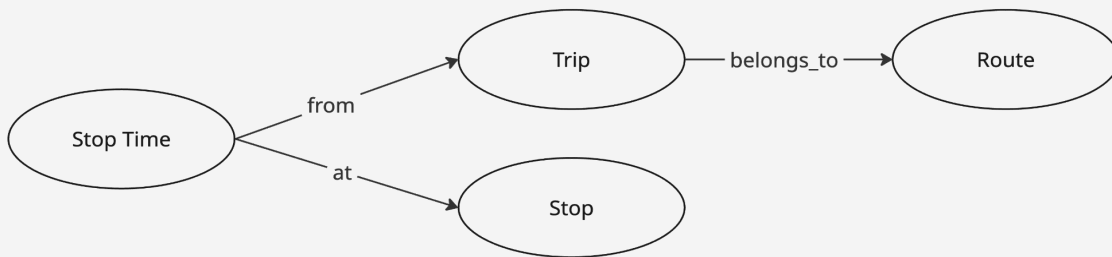
File: **Required**

Primary key ( route\_id )

| Field Name       | Type                                       | Presence                          | Description                                                                                                                                                                                                                                          |
|------------------|--------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| route_id         | Unique ID                                  | <b>Required</b>                   | Identifies a route.                                                                                                                                                                                                                                  |
| agency_id        | Foreign ID referencing<br>agency.agency_id | <b>Conditionally<br/>Required</b> | Agency for the specified route.<br><br>Conditionally Required:<br>- <b>Required</b> if multiple agencies are defined in <a href="#">agency.txt</a> .<br>- Recommended otherwise.                                                                     |
| route_short_name | Text                                       | <b>Conditionally<br/>Required</b> | Short name of a route. Often a short, abstract identifier (e.g may be defined).<br><br>Conditionally Required:<br>- <b>Required</b> if routes.route_long_name is empty.<br>- Recommended if there is a brief service designation. This : characters. |
| route_long_name  | Text                                       | <b>Conditionally<br/>Required</b> | Full name of a route. This name is generally more descriptive and route_long_name may be defined.                                                                                                                                                    |

# Ontology & Metadata

We kept the relationships and key metadata from the GTFS schema, just reduced in scope



| Type      | Field             | Description                                   |
|-----------|-------------------|-----------------------------------------------|
| route     | route_short_name  | route number ex. "3"                          |
|           | route_description | ex. "Capitol Hill - Downtown Seattle"         |
| stop      | stop_name         | ex. "1st Ave & Spring St"                     |
|           | stop_lat          | latitude                                      |
|           | stop_lon          | longitude                                     |
| trip      | route             | route node belonging to trip                  |
|           | trip_headsign     | readable identifier ex. "1st Ave & Spring St" |
| stop_time | trip              | trip node belonging to stop time              |
|           | stop              | stop node belonging to stop time              |
|           | stop_sequence     | relative position of stop in trip sequence    |
|           | arrival_time      | when bus arrives                              |
|           | departure_time    | when bus departs                              |

# Transformation of GTFS Format

1

## GTFS Files

Zipped .txt files,  
readable as CSVs

2

## Data Conversion

Reformatted CSV  
rows into RDF  
templates

3

## Construction

Constructed RDF  
knowledge graph in  
memory

4

## Serialization

Write knowledge  
graph into N-Triples  
format to share and  
import

5

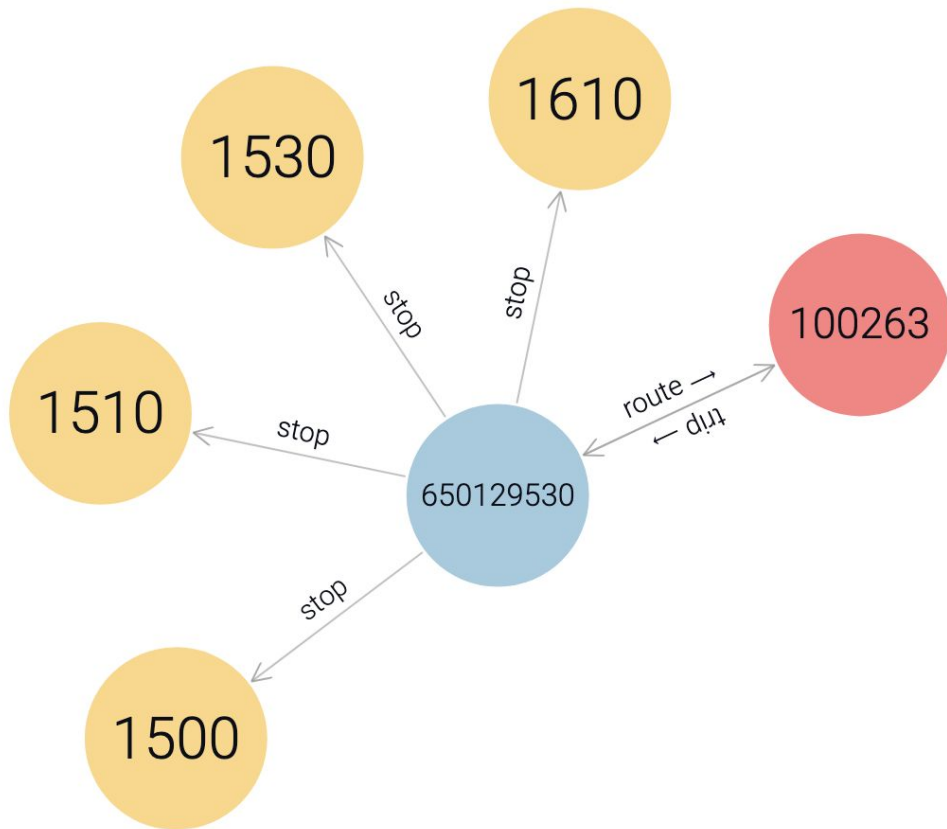
## Deployment

Import N-Triples file  
into graph database,  
or in memory

# Knowledge Graph: The core information structure

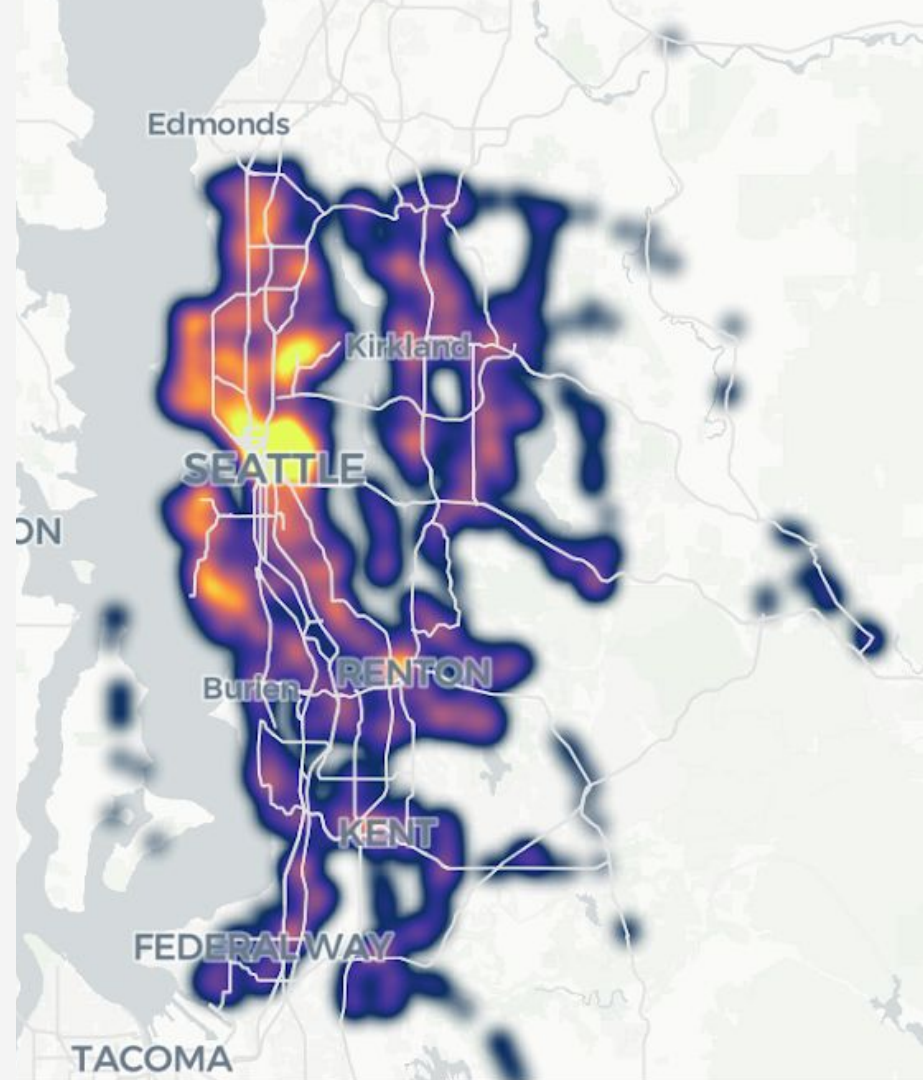
An RDF (Resource Description Framework) following Semantic Web Standards

Queryable via SPARQL endpoint



# Putting the Knowledge Graph to Work

Tasks like network analyses highlight the strengths of a graph data structure



Heat map of stops



# FAIR Assessment

1

## Findable

Our serialization retained the robust findability features of the original GTFS files: schema and metadata remain in place

2

## Accessible

N-triple file format is similar to JSON for human readability, while being efficiently machine ingestible

3

## Interoperable

N-triples are able to be ingested and displayed by a diversity of programs and databases, allowing for a wide variety of end user applications

4

## Reusable

N-triple formats does not lose anything from the original GTFS files, and can be converted back to the original .txt files, or converted to another file type with relative ease

# Test Plan

## Transit Lab Workflow

- Manually test github installation instructions
- Manually test Jupyter notebook (cell by cell or end to end)
- Check validity of output file

## Transit Lab Visualizer

- Unit test suite
- End to end tests from Streamlit front end to GraphDB back end
- Tracking initial query and start up times
- Queries can be tested in the GraphDB Workbench

# Performance and Availability

## Performance Issues:

- Time to Serialize
- Time to Query
- Size of queries

The screenshot shows the GitHub repository page for **IMT-542-Transit-Lab**, which is a public repository. The repository is currently on the **main** branch, with 2 branches and 0 tags. A pull request #1 is open, merged by **madison-abshire** 4 hours ago, with 20 commits. The repository contains several files and folders, including **archive**, **data**, **output**, **.gitignore**, **README.md**, **requirements.txt**, and **workflow.ipynb**. The **README** file is selected, showing the title **IMT 542 Transit Lab** and sections for **Introduction**, **Getting started**, and **Clone repository**. The right sidebar shows the repository's **About** section, including the owner **Lincoln Brennan/Madison Abshire**, a **Readme** link, **Activity**, **0 stars**, **0 watching**, and **0 forks**. The **Releases** section shows no releases published, and the **Packages** section shows no packages published. The **Contributors** section lists **lkbrennan97** and **madison-abshire**. The **Languages** section shows the repository is primarily composed of **Jupyter Notebook** files (93.8%) and **Dockerfile** files (6.2%).

IMT-542-Transit-Lab (Public)

main 2 Branches 0 Tags

Go to file Add file Code

madison-abshire Merge pull request #1 from lkbrennan97/madison 6c48c6f · 4 hours ago 20 Commits

|                  |                                                            |             |
|------------------|------------------------------------------------------------|-------------|
| archive          | archive previous project structure                         | last week   |
| data             | added data and sample zips                                 | last week   |
| output           | added data and sample zips                                 | last week   |
| .gitignore       | archive previous project structure                         | last week   |
| README.md        | Cleaned up workflow.ipynb and added set up instructions... | last week   |
| requirements.txt | remove unneeded requirements                               | 4 hours ago |
| workflow.ipynb   | removed gefx version                                       | 4 hours ago |

README

## IMT 542 Transit Lab

### Introduction

### Getting started

Clone repository

About

Lincoln Brennan/Madison Abshire

Readme

Activity

0 stars

0 watching

0 forks

Releases

No releases published

[Create a new release](#)

Packages

No packages published

[Publish your first package](#)

Contributors 2

lkbrennan97 Lincoln Brennan

madison-abshire

Languages

Jupyter Notebook 93.8%

Dockerfile 6.2%



Ask us anything!